TRACE-V: a PicoRV32 RISC-V core on SKY130, built by agentic AI and verified with independent open-source tools

Raisul Islam · October 2026 · Release v2.0.2

TRACE-V: Traceable RISC-V — AI-built, Checked, Evidenced. The release is published at `https://github.com/raisul1212/trace-v` for non-commercial academic instruction and research.

TRACE-V is a worked example of block-level implementation and verification. Reported signoff is limited to the stated checks, constraints and conditions. It is not fab-ready.

Release v2.0.2 contains two implementations of the same design, introduced in v2.0. The same RTL and the same constraint file were implemented by a commercial flow (Cadence Genus and Innovus) and by an open flow (Yosys and OpenROAD in LibreLane), and both builds are verified with the same open-source scripts. The commercial circuit artifacts originate in release v1.0.1; the verification scripts and the documentation were corrected in v2.0.2. Release v1.0.1 replaced the 8 ns route of v1.0 with an 8.25 ns route, which has a wider setup margin at the slow corner; the 8 ns release remains available at tag v1.0.

Abstract

- **Commercial build (v1.0.1).** The PicoRV32 RV32I core was taken from RTL to block-level signoff on SkyWater sky130A at 8.25 ns (about 121 MHz). The requirements were set and the decisions were approved by the designer; implementation, verification, debugging and fixes were carried out by agentic AI on a commercial implementation flow. The routed block has 8,273 cells in a 357.42 × 355.64 µm die.
- Setup slack is +0.180 / +2.388 / +2.957 ns and hold slack is +0.716 / +0.214 / +0.034 ns at ss / tt / ff, after 0.25 ns clock uncertainty. Slew, capacitance and fanout violations are 0 at all three corners.
- DRC (Magic and KLayout), antenna and LVS are clean, and both LEC steps are equivalent. The post-layout functional simulation of the golden program, with partial SDF timing (Icarus Verilog), passes at 8.25 ns and the 4.125 ns control fails; the CVC timing-check simulation released with v2.0.2 passes at 8.25 ns with 0 timing violations; a re-run in Xcelium with flop timing checks enforced also passes at 8.25 ns with 0 violations.
- Static IR drop is 11.0 mV on VPWR and 11.58 mV on VGND (values quoted from the Voltus run in the PSM comparison record) against the stated 80 mV budget; the released Voltus power report gives 12.98 mW at 20% primary-input activity.
- Six defects that each passed a tool's own report were found and corrected by agentic AI (section 9). Signoff is at nominal RC (section 11).
- **Open build (v2.0).** The same RTL, under the byte-identical constraint file, was implemented with Yosys and OpenROAD in LibreLane 3.0.14, with private modifications to OpenROAD that are not published (section 4). The die is 480.21 × 490.93 µm at 45% configured core utilization, with 144,143 transistors in logic cells.

- In the shipped OpenSTA run, setup slack is +0.129 / +2.028 / +2.706 ns and hold slack is +0.744 / +0.235 / +0.054 ns at ss / tt / ff, with 0 slew, capacitance and fanout violations. DRC (Magic and KLayout), antenna and LVS are clean. Functional simulation with partial SDF timing (Icarus Verilog) passes at 8.25 ns, and so does Xcelium with 0 timing violations; both controls fail. CVC, released with v2.0.2, passes at 8.25 ns with 0 timing violations; its 4.125 ns control fails functionally but with 0 timing-check violations, so it does not meet the control criterion. Both equivalence steps are equivalent in Conformal. Static IR from OpenROAD PSM is 5.554 mV on VPWR and 5.573 mV on VGND (section 5).
- Five defects found during the v2.0 work were fixed by agentic AI, among them a disagreement between two antenna checkers and a constraint mismatch (section 9.6).
- For each build, the release covers the following checks. Timing was signed off at three corners with 0.25 ns clock uncertainty. DRC was run with two independent decks, LVS was run against the routed netlist, and logic equivalence was checked at both synthesis and routing. Post-layout gate-level simulation was run with flop timing checks, and static IR was analyzed. The failure handling of the verify scripts is tested offline. Reported figures cite released evidence files with checksums; figures without a located report are withdrawn (see `docs/v2.0.2-errata.md`). The released package provides scripts and evidence for the checks listed in the verification table of its README. Icarus results establish functional simulation with partial SDF timing. CVC is the timing-check simulator of record; its released results and coverage are identified separately. Commercial-tool results (Conformal, Tempus, Innovus, Voltus, Xcelium) are released as scrubbed historical evidence and are not regenerated by the scripts; private open-tool development, implementation reconstruction and externally sourced background are outside the repository's reproduction guarantee.

Background: PicoRV32

PicoRV32 is a CPU core by Claire Xenia Wolf that implements the RISC-V RV32IMC instruction set. It can be configured as RV32E, RV32I, RV32IC, RV32IM or RV32IMC [1]. It is size-optimized, at 750–2,000 LUTs and 250–450 MHz on Xilinx 7-series FPGAs, and is intended mainly as an auxiliary processor in FPGA and ASIC designs [1]. Its average CPI is about 4, depending on the instruction mix. It is released under the ISC license [1]. The FPGA frequency, LUT count and CPI figures and the instruction-count background below are externally sourced background; they are outside this repository's evidence and reproduction guarantee, and they are not measurements of this ASIC package.

It is a general-purpose CPU. RV32I is RISC-V's base 32-bit integer instruction set: 40 distinct instructions covering integer arithmetic, logic, shifts, compares, loads, stores, branches, jumps and system [2]. M (multiply/divide) and C (compressed) are optional extensions [2].

This build uses PicoRV32's default parameters, unchanged [1]:

- RV32I with all 32 registers and the cycle and instruction counters;
- no multiply/divide, no compressed instructions, no interrupts and no co-processor interface;
- a non-barrel shifter;
- traps on misaligned accesses and illegal instructions.

The memory interface works as follows: the core asserts `mem_valid` and holds it until the memory asserts `mem_ready`. An optional look-ahead interface (`mem_la_*`) announces a transfer one cycle early [1].

The process is SkyWater SKY130, an open-source 130 nm PDK; this design uses the `sky130_fd_sc_hd` standard-cell library [3].

1 The design and target

Sections 1 to 3 describe the commercial build of v1.0.1; the open build is described in sections 4 to 6. The design is PicoRV32, an RV32I RISC-V CPU core by Claire Xenia Wolf, released under the ISC license. The process is SkyWater sky130A, an open PDK, with `sky130_fd_sc_hd` standard cells. The target is a single clock of 8.25 ns (about 121 MHz) with 0.25 ns of clock uncertainty applied at every corner, and a fanout limit of 10 that is enforced.

PicoRV32 here is a 32-bit RISC-V core (RV32I). The routed block holds 87,237 transistors in 8,273 standard cells. The count comes from the sky130 SPICE model of every placed cell; filler, well-tap and antenna-diode cells add none.

Corner	Temperature	Supply
ss	100 °C	1.60 V
tt	25 °C	1.80 V
ff	−40 °C	1.95 V

The final routed block of the commercial build has the following size and activity figures.

Quantity	Value
Die	357.42 × 355.64 µm
Cells	8,273 in the routed block, of which 176 are clock-tree buffers (7,845 in the synthesis netlist)
Cell area	75,681 µm^2 in the routed block (71,909 µm^2 in the synthesis netlist)
Clock-tree insertion delay	0.862–0.963 ns at ss, average 0.916 ns, rounded 0.86–0.96 and 0.92 ns (Innovus CTS statistics, `reports/innovus/`)
Power	12.98 mW, static (Voltus power report, `reports/voltus/`), 20% primary-input activity
Fanout limit	10, enforced

2 Flow and signoff bar

Implementation and timing signoff of the v1.0.1 build use a commercial flow. Cadence Genus does synthesis, Innovus does place and route, Tempus produces timing and the SDF, Conformal checks logic equivalence, and Voltus computes IR drop.

Physical signoff uses open checkers, so that anyone can re-verify the layout without the commercial tools: Magic DRC, KLayout DRC with the sky130A_mr deck, Magic antenna check, LVS with Magic and Netgen, and functional gate-level simulation in Icarus Verilog with partial SDF timing. Icarus Verilog applies SDF path delays only in part (interconnect annotations fail for some entries) and does not evaluate flop timing checks, so the simulation is repeated with the SDF timing checks enforced, in Cadence Xcelium and in the open-source CVC simulator (section 3).

The signoff bar was set once by the designer, and the work was carried out against it by agentic AI without renegotiation per run:

- 0.25 ns clock uncertainty at ss, tt and ff, included in both setup and hold.
- DRC clean in two independent decks.
- Max transition, max capacitance and max fanout violations at 0.
- Gate-level simulation and both LEC steps pass.
- Static IR drop within budget, 80 mV on each rail (5% of 1.6 V).

The open build of v2.0 was checked against the same bar (sections 4 and 5).

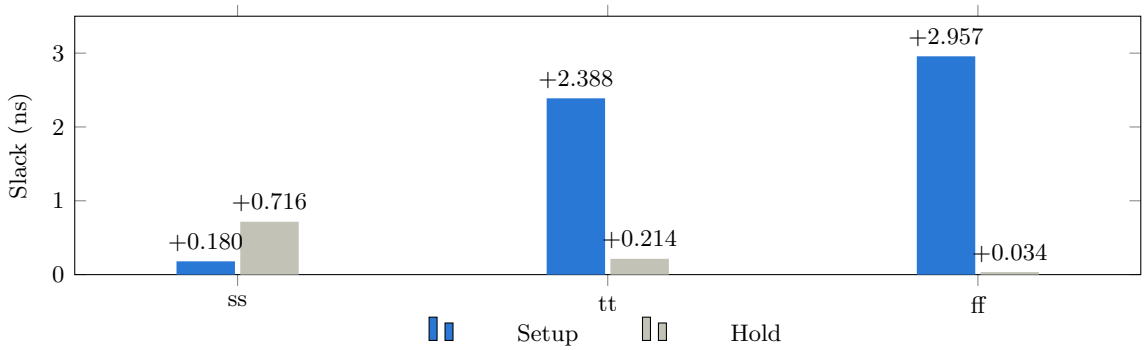
The workload for simulation is a hand-assembled RV32I golden program of 9 instructions that sums 1 to 10 and stores 55.

3 Results: commercial build

Every check below passes at 8.25 ns for the commercial build. Setup and hold both include the 0.25 ns clock uncertainty.

Check	Result
DRC, Magic	0 violations
DRC, KLayout	0 findings
Process antenna, Magic	0 violations
LVS, Magic + Netgen, against the routed netlist	Matched; 8,492 / 8,492 nets
LEC, RTL → synthesized netlist (Conformal)	Equivalent; 1,916 compare points, 0 non-equivalent, aborted or unknown
LEC, synthesized → routed netlist (Conformal)	Equivalent; 1,864 compare points, 0 non-equivalent, aborted or unknown
Setup slack, ss / tt / ff (Tempus, `reports/tempus/`)	+0.180 / +2.388 / +2.957 ns; 0 violating endpoints
Hold slack, ss / tt / ff (Tempus)	+0.716 / +0.214 / +0.034 ns; the ss report covers 1,626 reported endpoints, none violating; this is not a claim that every interface is constrained
Max transition / capacitance / fanout	0 / 0 / 0 at all three corners
Golden program on RTL	Pass
Golden program, post-layout functional simulation with partial SDF timing (Icarus), 8.25 ns	Pass; the 4.125 ns delay-sensitivity control fails
Gate-level simulation with timing checks (Xcelium), 8.25 ns	Pass, 0 violations; the 4 ns control fails (76 violations)
Timing-check simulation with CVC (released, 2026-10-16), 8.25 ns	Pass, 0 timing violations; 20,796 of 20,796 IOPATH, 22,090 of 22,090 INTERCONNECT and 6,228 of 6,228 TIMINGCHECK entries applied; all 9,342 timing checks given limits; the 4.125 ns control fails with 63 timing-check violations
CVC at other periods, same netlist and SDF (released)	8.0 ns: pass, 0 violations; 6.0 ns: fail (RESULT timeout, 37 violations). Two periods on one workload
Static IR drop, VPWR / VGND	11.0 mV / 11.58 mV (quoted from the Voltus run in the PSM comparison record); stated budget 80 mV each (5% of 1.6 V)

Setup and hold slack per corner: ss setup +0.180 ns, ff hold +0.034 ns
Slack in ns at 8.25 ns, after 0.25 ns clock uncertainty



Setup and hold slack per corner at 8.25 ns, including 0.25 ns clock uncertainty; values from the signoff results table

Each simulation result has a control. The Icarus 4.125 ns control checks delay sensitivity for this workload; its failure does not establish complete SDF annotation. Icarus provides functional simulation with partial SDF timing and does not evaluate the required timing checks: the shipped commercial logs report about 5,080 failed interconnect annotations and 1,557 unsupported TIMINGCHECK warnings per run, and the open logs report 196 and 1,597. Annotation coverage is measured with CVC, the timing-check simulator of record, on the simulation SDF (the 454 port-endpoint INTERCONNECT entries removed from the full SDF are not timed): all IOPATH and TIMINGCHECK entries and all INTERCONNECT entries were applied, and CVC at 8.25 ns reports 0 timing violations while its 4.125 ns control reports 63. At 8.0 ns CVC reports 0 violations; at 6.0 ns the program times out with 37 violations. These are two further periods on one workload and do not locate a general timing limit.

Post-layout gate-level simulation was re-run in Cadence Xcelium with flop timing checks enforced, using the netlist, SDF and testbench of the v1.0.1 harness; their hashes are recorded in `reports/xcelium/summary.json` and are not byte-identical to the cleaned files shipped here. The annotation table of each log reports all 3,114 setup/hold and 3,114 pulse-width entries annotated (100%). At 8.25 ns the program passes with 0 timing-check violations, and at 10 ns it passes with 0. At 4 ns, the control, it times out with 76 violations. The counts are the timing-violation diagnostic blocks of the released logs; the job's own summary lines (2, 2 and 154) are a grep artefact.

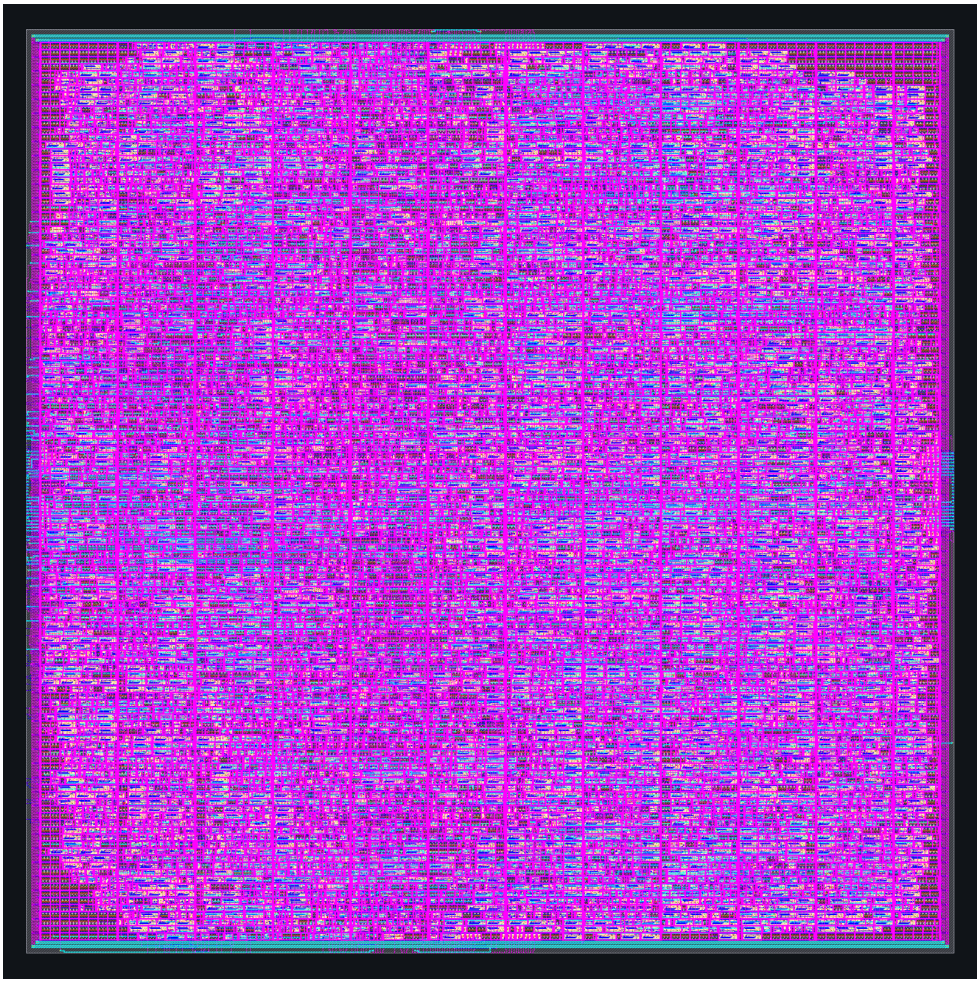


Figure 1. Commercial build: routed PicoRV32, 357.4 × 355.6 µm, sky130A (KLayout render of the final GDS).

4 Open-flow implementation

Release v2.0 adds a second implementation of the same RTL under the same constraint file. The file `constraints/picorv32.sdc` is byte-identical in both packages (sha256 prefix `286822dc54798fb9`). It sets an 8.25 ns clock with 0.25 ns uncertainty, a maximum fanout of 10 and a half-period I/O delay of 4.125 ns, because the testbench drives on the falling edge. The outputs `mem_la_read`, `mem_la_write`, `mem_la_addr[*]`, `mem_la_wdata[*]`, `mem_la_wstrb[*]`, `pcpi_valid`, `pcpi_insn[*]`, `pcpi_rs1[*]`, `pcpi_rs2[*]`, `eoi[*]` and `trace_*` have no output-delay constraints. The PCPI inputs `pcpi_wr`, `pcpi_rd[*]`, `pcpi_wait` and `pcpi_ready` have 4.125 ns input delays.

The open build was produced with LibreLane 3.0.14 in its Classic flow:

- Yosys synthesis, with `SYNTH_ABC_BUFFERING` on and the `DELAY 0` strategy;
- OpenROAD placement and routing;
- configured core utilization of 45%, a clock-tree buffer spacing of 100 µm, a maximum fanout of 10, an antenna repair margin of 20 and a clock period of 8.25 ns (the recorded configuration,

`reports/implementation/config.json`; these are settings, not measured outcomes);

- reports supporting failures at 55% or 65% utilization, or 16 root-buffer loads at 40 to 45%, were not located; those statements are withdrawn;
- two antenna diodes inserted after routing with LibreLane's stock `Odb.InsertECODiodes` step (section 9.6).

Equivalent tools

Step	Commercial build (v1.0.1)	Open build (v2.0)
Logic synthesis	Cadence Genus 25.11	Yosys 0.62 with ABC, inside LibreLane 3.0.14
Floorplan, placement, clock tree, routing, timing repair	Cadence Innovus 25.11	OpenROAD (upstream `dcf36133`) with private modifications, driven by LibreLane 3.0.14
Parasitic extraction for timing	Innovus, with the typical extraction file of the Cadence sky130 kit (nominal RC)	OpenRCX, inside OpenROAD
Signoff static timing	Cadence Tempus 23.10	OpenSTA inside OpenROAD (upstream `857316ff`), with private modifications
Static IR drop	Cadence Voltus	OpenROAD PSM with a private modification
Antenna repair	Innovus NanoRoute with an antenna margin	OpenROAD antenna repair, then two diodes added with LibreLane's stock `Odb.InsertECODiodes` step
Logic equivalence	Cadence Conformal 24.10	Cadence Conformal 24.10 (same tool)
Gate-level simulation with timing checks	Cadence Xcelium 24.03	Cadence Xcelium 24.03 (same tool)
Release verification (`verify/`)	Magic 8.3.623, KLayout 0.30.7, Netgen 1.5.316, OpenSTA 2.7.0, Icarus Verilog 13.0	the same unmodified tools

Modifications inside the open-source tools

Implementation used OpenROAD with private modifications that are not published. Every check an outsider runs uses unmodified open-source tools. The modifications are in four places.

1. **OpenROAD Resizer (`rsz`): electrical repair and rebuffering.** After routing, repair uses the extracted parasitics, including zero-length shunt capacitances, in place of an estimate that had under-counted wire load by 2.2 to 2.4 times on the nets of a parasitic probe (a development record that is not released here). Driver sizing uses the selected corner's model and the SDC slew limit, and fanout headroom accounts for antenna-diode loads.
2. **OpenSTA inside OpenROAD: parasitics and SDC.** The SPEF reader and the parasitic store give the Resizer the extracted RC network in a form its buffering model can use, and constraint checks ignore empty exception entries.
3. **OpenROAD PSM: static IR.** Voltage sources can be placed on a power pin's own routing

layer as well as on the top layer. With this change, the PSM result on the commercial layout is 12.09 / 12.07 mV against the Voltus values 11.0 / 11.58 mV quoted in the comparison record; this does not establish general agreement between the tools.

4. **LibreLane: flow integration.** A post-route loop repairs electrical, setup and hold violations against freshly extracted parasitics, the flow exits with failure when a stated setup target is missed, and unchanged detailed wires are preserved.

Yosys and ABC, Magic, KLayout, Netgen and Icarus Verilog were not modified. The antenna repair of the final layout uses a stock LibreLane step.

Quantity	Value
Die	480.21 × 490.93 µm
Standard-cell area	122,928 µm^2, of which 3,919 µm^2 are tap cells (119,009 µm^2 without them)
Cells	10,801 logic cells, plus 3,132 taps, 494 antenna diodes, 19,498 decap and 11,766 fill
Transistors	183,139, of which 144,143 are in logic cells and 38,996 in decap cells
LVS size	11,237 devices and 10,944 nets

The transistor count uses the method of the shipped `verify/transistors.sh`: the DEF components multiplied by the transistors per cell in the PDK cell SPICE. The commercial build has no decap cells, so its 87,237 transistors are all in logic cells. The open build is reported to have been implemented with open_pdks `8afc8346`, the version LibreLane 3.0.14 selects; that is reported from private runs and no released environment record establishes it. A readback of two PDK installs on 2026-10-16 found the cell LEF, the ss and tt timing libraries and `primitives.v` byte-identical, and the ff timing library and the `sky130_fd_sc_hd.v` cell model different (`reports/environment/`). Both builds were verified with `97d08448`.

5 Open-flow verification

Every result in this section was measured on the final open layout. Timing, DRC, antenna, LVS and the Icarus Verilog simulation come from a run of the shipped open-source scripts; the other results name their tool.

Timing

OpenSTA was run with the shipped `verify/sta.tcl`, the nominal SPEF and the shared SDC. Slew, capacitance and fanout violations are 0 / 0 / 0 against library limits.

Corner	Setup slack	Hold slack
ss	+0.129 ns	+0.744 ns
tt	+2.028 ns	+0.235 ns
ff	+2.706 ns	+0.054 ns

The released OpenSTA evidence covers ss, tt and ff using the same nominal-RC SPEF. Nine-corner signoff is reported from private open-tool runs, not released in this repository.

Physical checks

- **DRC.** Magic reports 0 errors, and KLayout with the sky130A_mr deck reports 0 findings.
- **Antenna.** Magic `antennacheck` reports 0 violations, with 30,900 gates analyzed.
- **LVS.** Netgen, against a Magic extraction from the GDS, reports that the circuits match uniquely, with 11,237 devices and 10,944 nets.

Simulation

Icarus Verilog, as functional simulation with partial SDF timing, passes at 8.25 ns (RESULT 55, VERDICT PASS), and the 4.125 ns delay-sensitivity control fails (RESULT timeout). CVC with the SDF timing checks, run on 2026-10-16 on the shipped simulation SDF, passes at 8.25 ns (RESULT 55) with 0 timing violations: 26,463 of 26,463 IOPATH and 6,388 of 6,388 TIMINGCHECK entries were applied, 25,989 of 26,002 INTERCONNECT entries were applied with 13 zero-delay entries skipped, and all 9,582 timing checks were given limits. Its 4.125 ns control fails functionally (RESULT timeout) with 0 timing-check violations, so by the control criterion (a failure with at least one timing-check violation) the open build's CVC control is not a pass; it shows delay sensitivity of the workload, not a live timing check. The open simulation SDF retains 70 port-origin INTERCONNECT entries, including the clock.

Xcelium 24.03 was run with the SDF and flop timing checks (`-neg_tchk`), in the same harness as the v1.0.1 run. The annotation table of each log reports 6,388 timing-check entries (3,194 setup/hold and 3,194 pulse-width) and 76,271 path delays annotated (100%); the logs are in `reports/xcelium/`.

- At 8.25 ns the program passes (RESULT 55, VERDICT PASS) with 0 timing violations.
- At 10 ns it passes with 0 violations.
- At 4 ns, the control, it fails (RESULT timeout) with 2 timing violations.

For comparison, the commercial build has 0 violations at 8.25 ns with 3,114 setup/hold and 3,114 pulse-width entries annotated, and its 4 ns control has 76 violations. The counts come from the diagnostic blocks of the logs; the job summaries' counts (2, 2 and 6 here) are a grep artefact.

Equivalence

Conformal LEC 24.10 was used for both steps.

- **RTL → Yosys netlist: equivalent.** There are 1,959 compare points (1,862 compared points and 97 reported equivalences), with 0 non-equivalent, 0 aborted and 0 unknown (`reports/conformal/`).
- **Yosys netlist → routed netlist: equivalent.** There are 1,904 compare points, with 0 non-equivalent and 0 aborted.

The RTL-to-netlist comparison required three steps:

1. flip-flop instances were renamed after their RTL registers, by a script that asserts identical cells and connections;
2. FSM encoding files were given for the two state registers that Yosys re-encoded: `cpu_state` from 8-bit one-hot to 7-bit one-hot, and `mem_wordsize` from 2-bit binary to 3-bit one-hot;
3. Conformal setup analysis merged 77 RTL registers (duplicates and registers the default configuration never uses) and treated 5 as constants.

The control is the same comparison without the encoding files, in which 1,246 points are non-equivalent. A stage-1 proof aid without setup analysis (not the shipped design) leaves 1,202. Both numbers rest on the setup record in `reports/conformal/`; the logs of these two controls were not released. They are diagnostic configurations, distinct from the final equivalence checks.

Static IR

Static IR was computed with OpenROAD PSM, with private modifications, on the final layout (`reports/psm/u45eco2/`); the total power is 22.0 mW. The released job files do not state the corner, the input activity or the supply-source placement, so the earlier statement of 8 supply pins per net on met4/met5 is not carried over; the same method on the commercial layout used the ss 100 °C, 1.6 V liberty at 20% input activity.

- The worst drop is 5.554 mV on VPWR and 5.573 mV on VGND. The average is 2.85 mV on VPWR and 2.75 mV on VGND.
- The stated budget is 80 mV, 5% of 1.6 V.
- The controls give a single source 32.2 mV, resistance multiplied by 3 16.7 mV, and resistance multiplied by 25 138.9 mV, which is above the stated budget (the first two are below it). They show the responses of these runs; they do not show that every verification wrapper enforces the budget.
- On the commercial layout the same method gives 12.09 / 12.07 mV, against Voltus values of 11.0 / 11.58 mV quoted in the comparison record (`reports/psm/`); this does not establish general agreement between the tools.

No Voltus run exists on the open layout.

6 The two builds side by side

	Commercial (v1.0.1)	Open (v2.0)
Clock / constraints	8.25 ns, identical SDC	8.25 ns, identical SDC
ss setup (shipped OpenSTA, nominal RC)	+0.280 ns	+0.129 ns
ff hold (shipped OpenSTA, nominal RC)	+0.032 ns	+0.054 ns
Die	357.42 × 355.64 µm	480.21 × 490.93 µm (45% configured utilization)
Logic transistors	87,237	144,143
DRC / antenna / LVS	0 / 0 / match	0 / 0 / match
Static IR (worst)	11.0 / 11.58 mV (Voltus, quoted)	5.55 / 5.57 mV (PSM)
Equivalence	RTL → netlist → routed (Conformal)	RTL → netlist → routed (Conformal)

The IR figures come from different tools. The PSM method, run on the commercial layout, gives 12.09 / 12.07 mV. This does not establish general agreement between the tools.

Step by step: tool and result in each flow

Step	Commercial: tool, result	Open: tool, result
Synthesis	Genus 25.11; 8,273 cells in the final layout (7,845 after synthesis)	Yosys 0.62 with ABC; 10,801 logic cells in the final layout
Place and route at 8.25 ns	Innovus 25.11; closes	OpenROAD, modified; closes (a development record of a stock-OpenROAD run on the reference configuration with the default SDC reports −2.053 ns at max_ss with 3,104 slew violations; that record is not released here)
Timing, shipped OpenSTA script	ss setup +0.280 ns, ff hold +0.032 ns	ss setup +0.129 ns, ff hold +0.054 ns
Static IR, worst VPWR	Cadence Voltus; 11.0 mV (quoted)	OpenROAD PSM, modified; 5.55 mV. On the commercial layout: 12.09 mV (unmodified PSM: 46.4 mV)
Logic equivalence (commercial tool in both)	Conformal 24.10; RTL → netlist → routed proven	Conformal 24.10; RTL → netlist → routed proven
Simulation with timing checks (commercial tool in both)	Xcelium 24.03; 0 violations, 6,228 timing-check entries	Xcelium 24.03; 0 violations, 6,388 timing-check entries
DRC / antenna / LVS (open tools in both)	Magic, KLayout, Netgen; 0 / 0 / match	Magic, KLayout, Netgen; 0 / 0 / match

Conformal and Xcelium were used for the open build as well. The open equivalence method did not prove the RTL-to-netlist step for this design (the recorded attempt proved 586 of 1,864 compare points and left 1,278 unproven), and Icarus Verilog does not evaluate timing checks (CVC, released, does). The open build's implementation uses no commercial tool.

7 Software developed for this work

The design was implemented and verified by an agentic chip-design framework under the designer's direction. The software developed for it falls into three groups.

1. **Modifications inside open-source tools**, the four listed in section 4. They are private and not published.
2. **Adapters of the agentic framework.** An adapter drives one tool step, reads its settings back, refuses a run it cannot verify, and records the result.
   - Used for this release, commercial flow: adapters for Genus synthesis, Innovus place and route, Tempus timing, Voltus static IR and Conformal logic equivalence, behind an evidence registrar that writes each result to a ledger and freezes the run with checksum manifests. The Xcelium simulation ran as a recorded job.
   - Used for this release, open flow: LibreLane runs with recorded configurations and hashes; the Conformal equivalence set-up (flip-flop renaming checked for identical structure, and FSM encoding files); PSM runs with a source-feed control and a weakened-grid control; and the stock-LibreLane ECO diode run.
   - Used for both builds: the verification scripts shipped in each folder's `verify/`.
   - Developed alongside: an open logic-equivalence adapter over Yosys and ABC, whose run on this design is reported only as the negative result in the step-by-step table (586 of 1,864 compare points proven; it did not prove the RTL-to-netlist step); and, not the source of any released number, an open static-IR adapter over OpenROAD PSM and a LibreLane runner and pre-flight plugin that read back every setting.
3. **Evidence handling.** Results were recorded in a ledger and the runs were frozen with checksum manifests (the eight historical manifests are in `commercial/reports/historical_inventory/`). The failure handling of the verify scripts is tested offline by `tests/test_verify_failures.py`. The historical damaged-input controls known from the released reports show the stated responses in those runs; they do not establish that every checker detects every damaged input.

8 How the platform works

The requirements were set and the decisions were approved by the designer. Implementation, verification, debugging and fixes were carried out by agentic AI against the signoff bar fixed in section 2. Five properties determine whether the results can be trusted.

1. **Numbers cite their source.** Figures cite the run that produced them, and the released evidence files carry provenance records (original and published checksums). A figure for which no report was located is withdrawn, not quoted from memory or from a log summary.
2. **Historical runs are frozen with checksums.** A reconstruction from eight historical manifests gives 75 entries: 66 in the six implementation and verification sets plus Voltus, and 9 in the Xcelium set. A checksum check of the original files on 16 October 2026 checked 75 entries with 0 mismatches. This is a reconstruction and a new readback, not a total that the project stated at the time, and it counts manifest entries, not distinct contents.

3. **Key checks have controls.** A check is better trusted after it has been run on a known-bad input. Gate-level simulation fails at 4 ns. The antenna check fails on a frozen known-bad layout. The slew, capacitance and fanout check flags a deliberately overloaded net (a 5 pF load; ss 1 / 0 / 7 and tt, ff 1 / 0 / 5 capacitance, fanout and transition violations). A check that passes every input is treated as broken. These controls show the stated responses in those runs; they do not establish that every checker detects every damaged input.
4. **Refusal over silence.** A check that cannot run, or whose output cannot be read, is counted as a failure and never as a pass. The scripts clear earlier results, check every tool status and require a fresh single-line verdict; `tests/test_verify_failures.py` tests this offline. That test shows that failures are reported as failures, not that a design is correct.
5. **Independent checkers.** The tool that builds something is always checked by at least one other tool, because a tool checking its own work shares its own blind spots. Innovus builds the layout, but DRC is run by Magic and by KLayout, and LVS by Magic and Netgen. LVS proves the layout matches the routed netlist; Conformal LEC then proves the routed netlist equals Genus' netlist, and Genus' netlist equals the RTL. The result is an unbroken chain of independent checks from the GDS back to the source code. In the open build, OpenROAD builds the layout, and the same open checkers and Conformal verify it.

9 What the platform caught

Each of the six defects below, found in the commercial flow, passed its tool's own report: the flow finished and the tool reported success. For each defect, the symptom, how it was found, the root cause and the fix are given. Three practices exposed the defects:

- **Read back.** Each result was read back from the design and compared with the requirement.
- **Cross-checked.** Results that should agree were compared.
- **Controlled.** Key checks were run on a known-bad input.

9.1 The fanout limit was declared and ignored

- **Symptom.** The SDC set a maximum fanout of 10, yet Innovus finished with a fanout violation. The tool's flow completed normally.
- **How found.** It was flagged by the slew, capacitance and fanout check. That check is trusted because it flags a deliberately overloaded net (a 5 pF load).
- **Root cause.** The router's optimizer and the clock-tree builder each need fanout repair enabled explicitly.
- **Fix.** With repair enabled in both, the violation is gone. The clock tree grew and its insertion delay became longer. The numerical before and after figures given in v2.0.1 (violation count, buffer count, insertion delay) are withdrawn because no supporting report was located; the final clock-tree statistics are in `reports/innovus/`.

9.2 STA passed while the gate-level simulation trapped

- **Symptom.** After the fanout fix, setup was met at 8 ns, but the post-layout simulation hit a trap at 8 ns.
- **How found.** The failing transaction that occurred earliest in the run was traced by agentic AI. The core misread an instruction it had been given correctly. The interface was then timed: the testbench drives memory on the falling edge, so `mem_ready` has half a cycle, and the path `mem_ready` → `reg_op1` had negative slack (the earlier value is withdrawn: no report located).
- **Root cause.** The constraints set no input or output delays, so STA gave those paths a full cycle. The larger clock tree had made them longer.
- **Fix and numbers.** The SDC now states the environment the testbench models, with half-period input and output delays. The simulation then passes at 8 ns, and the input-port hold violations are gone (the earlier slack values are withdrawn: no report located).

9.3 The gate-level simulation was all-X

- **Symptom.** Every simulation failed, at every clock.
- **How found.** A zero-delay positive control: the same netlist with no delays must pass, and it did not.
- **Root cause.** The power-aware cell models had unconnected supplies, so every output resolved to X.
- **Fix.** The simulated netlist now ties every cell's supply and body pins (VPWR, VGND, VPB, VNB) to constant 1 or 0 before simulation. A zero-delay run of the same netlist must pass; that positive control would have caught the all-X result.

9.4 The antenna check could not fail

- **Symptom.** The initial negative control for the antenna check passed. It contained a diode that was not needed, so it was not a violation, and the check had not been shown able to fail.
- **Root cause.** The control was not a known-bad input.
- **Fix.** It was replaced with a frozen known-bad layout that must fail. The final design has 0 antenna violations.

9.5 A cell excluded from synthesis was re-inserted by place and route

- **Symptom.** KLayout reported one npc.2 violation on the initial 8.25 ns route, while Magic DRC was clean (the spacing values given in v2.0.1 are withdrawn: no report located).
- **How found.** The violation was reported by the second, independent DRC deck.
- **Root cause.** The cell `sky130_fd_sc_hd__and2_0` has an NPC edge inside its boundary. It was excluded in Genus, but restructuring in Innovus re-inserted it.
- **Fix.** The exclusion is applied in Innovus as well, and a post-route count of excluded cells must be zero.

9.6 An antenna violation remained after the router's repair

- **Symptom.** One net had an antenna ratio of 403.85 against a limit of 400 (the Magic antenna log of a development run, `commercial/reports/development/`).
- **How found.** The violation was reported by the open antenna check.
- **Root cause.** The router's repair did not bring the net below the limit.
- **Fix.** The router's antenna repair setting was changed (the 96% figure given in v2.0.1 is withdrawn: no report located).

Defects found and fixed during the v2.0 work

The same practices were applied to the open flow. Five defects were found and fixed by agentic AI. The numerical values given for these defects in v2.0.1 have no located supporting report except where a file is cited; the others are withdrawn.

Constraint mismatch

- **Symptom.** The open flow initially used LibreLane's default SDC, with an I/O delay that differed from the commercial SDC's.
- **How found.** The build was timed under the commercial SDC, where it failed ss setup at the input `resetn`.
- **Fix.** The build was re-run under the identical SDC, and it closes.

Fragile closure

- **Symptom.** Before the timing-repair modification, an 8.25 ns closure failed after a small placement change (a change of the wire-length weight).
- **Fix.** After the modification, the same perturbation passed.

Antenna checker disagreement

- **Symptom.** OpenROAD reported 0 antenna violations, and Magic flagged gates on the candidate layouts.
- **Root cause.** Magic counts metal3 that reaches a gate through metal4 alone as connected at the metal3 step.
- **Evidence.** The net-level ratio comparison and the copy-of-layout experiment behind this diagnosis have no located report and are withdrawn; the diagnosis is the project's explanation, not released evidence.
- **Fix.** On the released 45% layout, two diodes were added with a stock ECO step (the recorded configuration has two ECO diode targets). The shipped DEF holds 494 diodes (`open/reports/development/final-diode-count.json`); this does not establish a 492 to 494 history, and the area and slack changes given in v2.0.1 are withdrawn.

- **Alternative considered and not used.** A larger patched mode (its figures are withdrawn: no report located).

Verification environment

- **Symptom.** With a newer PDK than the pinned one, the diode cells failed LVS.
- **Root cause.** LVS and antenna results depend on the PDK version.
- **Fix.** The README pins open_pdks `97d08448`, and the scripts must be run with it.

Equivalence setup

- **Symptom.** Re-encoded state machines and constant registers defeat a plain key-point comparison: 1,246 non-equivalent points without the FSM encoding files, and 1,202 in the stage-1 plain run, until they were handled (setup record, `open/reports/conformal/`).
- **Fix.** The three steps in section 5 (register renaming, FSM encoding files and Conformal setup analysis) resolve it.

10 Power and IR: commercial build

For the commercial build, the static IR drop from Voltus is 11.0 mV on VPWR and 11.58 mV on VGND; these values are quoted from the Voltus run in the PSM comparison record, and the released Voltus report (`reports/voltus/`) is a power report that does not itself show IR-drop values. The stated budget is 80 mV on each rail, 5% of 1.6 V, so the block uses about 14% of the budget on VPWR. The legend of the VPWR map spans 1.588 to 1.600 V. The power report gives 12.98 mW, static, with 20% primary-input activity.

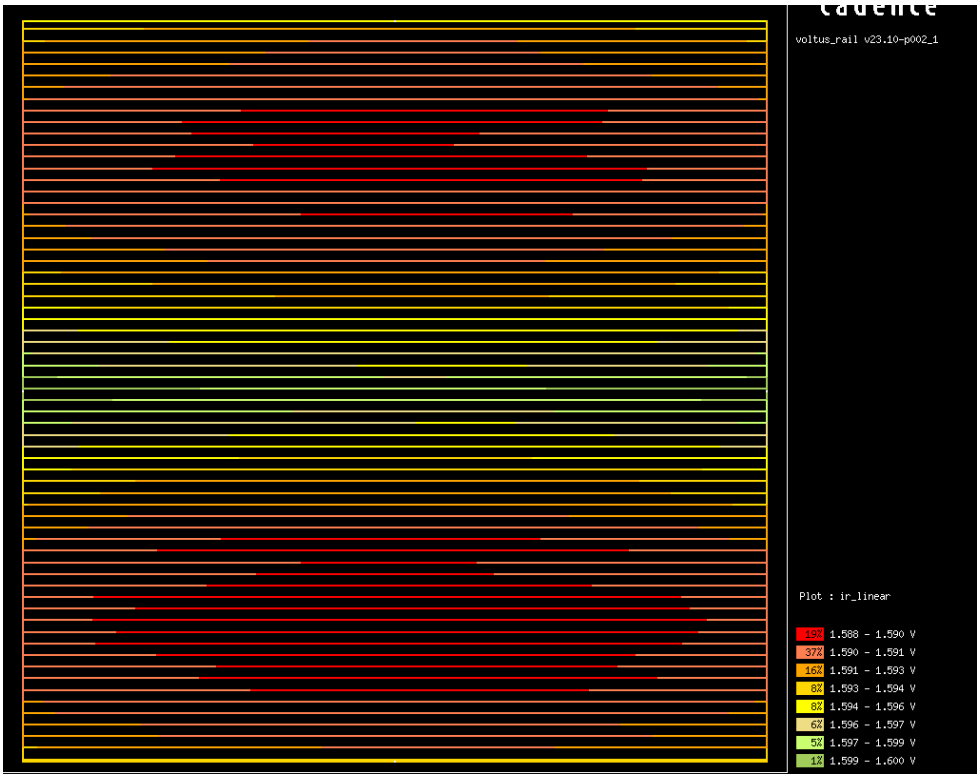
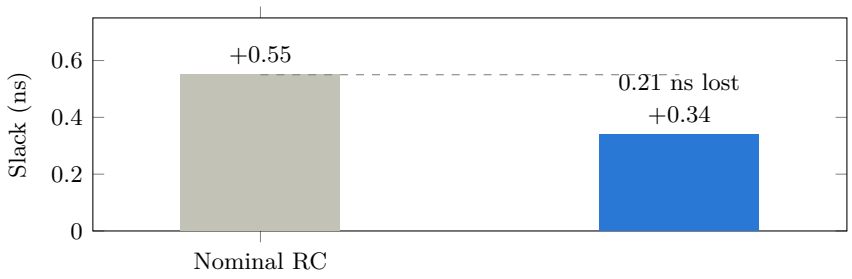


Figure 2. Commercial build: static IR map of VPWR (Voltus, vectorless): legend 1.588–1.600 V, worst drop 11.0 mV (quoted) against the stated 80 mV budget.

11 Independent RC cross-check: commercial build

Signoff extraction of the commercial build is at nominal RC, because the Cadence sky130 kit ships a typical extraction file and no other corner. To bound what that leaves out, the final layout was re-extracted with OpenRCX and timed with OpenSTA, tools independent of the signoff flow.

Max RC costs 0.21 ns of ss setup slack in the OpenRCX and OpenSTA check
ss setup slack at 8.25 ns, in ns; bars start at 0



ss setup slack at 8.25 ns, nominal against max RC; OpenRCX re-extraction timed in OpenSTA

ss setup at 8.25 ns is +0.55 ns with nominal RC and +0.34 ns with max RC, so max RC costs 0.21 ns. The ff hold check with min RC is +0.03 ns and is met.

OpenSTA on the Innovus nominal SPEF gives +0.28 ns in the same record (`reports/rc_crosscheck/`). Subtracting the 0.21 ns from the separate Tempus margin of +0.180 ns gives −0.030 ns, and adding that deficit to the 8.25 ns period gives 8.28 ns, about 8.3 ns, if period and slack change one for one. This is a heuristic that combines results of different analyses; it is not a measured max-RC timing requirement. The shipped three-corner checks still reuse one nominal-RC SPEF.

12 Cost

The accounting window is 1 to 7 October 2026, ending at 16:15:04 EDT. It records 2,907 calls and the following tokens: 6,524 uncached input, 1,261,596 output, 31,934,690 cache-creation and 891,055,522 cache-read. Interaction accounting records 86 messages, 3.14 hours under the 20-minute rule, and 7 days. The scheduler dataset contains 62 jobs with 5.14 hours (5.1386 h) of summed job elapsed time; this is not CPU-hours and not the elapsed duration of the project. The identified seven-job implementation pass totals 37 min 35 s of job elapsed time over a 42 min 56 s span. The totals can include concurrent work outside this design and are an upper-bound attribution. The data and method are in `commercial/reports/accounting/`. The claim of 21 further jobs and 2.0 hours is withdrawn because its accounting data was not located.

13 Limitations

Packaging, electromigration, metal fill, pad frame, foundry precheck and dynamic IR were outside the scope of this block-level release. Items 1 and 3 apply to the commercial build; items 2 and 4 apply to both builds; items 5 and 6 apply to the open build.

1. **Nominal RC.** Signoff is at nominal RC (section 11).
2. **I/O timing.** I/O timing is set to the testbench's half-cycle memory protocol. The outputs `mem_la_read`, `mem_la_write`, `mem_la_addr[*]`, `mem_la_wdata[*]`, `mem_la_wstrb[*]`, `pcpi_valid`, `pcpi_insn[*]`, `pcpi_rs1[*]`, `pcpi_rs2[*]`, `eoi[*]` and `trace_*` have no output-delay constraints; the PCPI inputs `pcpi_wr`, `pcpi_rd[*]`, `pcpi_wait` and `pcpi_ready` have 4.125 ns input delays.
3. **Thin margins.** The margins are ss setup +0.180 ns and ff hold +0.034 ns, after 0.25 ns uncertainty.
4. **Workload.** The post-layout simulation runs one 9-instruction program covering 6 of 40 RV32I opcodes. Logic equivalence concerns the configured design (default parameters). STA covers the paths constrained by the shipped SDC; it does not establish timing coverage for every interface. CVC times the simulation SDF, which lacks the port-endpoint INTERCONNECT entries, on one workload.
5. **Open-build IR.** The static IR of the open build comes from OpenROAD PSM with private modifications. No Voltus run exists on the open layout; the PSM method was validated on the commercial layout.

6. **Commercial checks of the open build.** The equivalence results (Conformal) and the Xcelium timing-checked simulation of the open build use commercial tools and are not regenerated by the open scripts; the timing-checked simulation is reproduced with CVC.

14 Reproducing

The repository at `https://github.com/raisul1212/trace-v` contains two self-contained packages, `commercial/` and `open/`, each with its own verify/ scripts, reports and SHA256SUMS. Each package contains:

- rtl and netlists;
- layout: the GDS, byte-identical to the one used at signoff, and the DEF;
- constraints, SPEF and SDF;
- the testbench;
- verify/ scripts for the open-source tools;
- the reports from the open-tool verification run (the v2.0.1 run; v2.0.2 changed the scripts and did not repeat it), and the CVC, area and library-comparison results newly run on 2026-10-16;
- scrubbed historical evidence from the original runs under `reports/` (Conformal, Tempus, Innovus, Voltus, Xcelium, PSM, RC cross-check, accounting, historical manifests), each directory with a `provenance.json`;
- SHA256SUMS.

The repository root holds the README, LICENSE, NOTICE, LICENSES/ and docs/ with the presentation and whitepaper PDFs.

The physical checks of each package can be re-run with the open-source tools and the scripts in its verify/, inside the LibreLane 3.0.14 container, with open_pdks `97d08448` installed by volare. LVS and antenna results depend on the PDK version, so the pinned PDK is required. The commercial-tool signoff figures (Genus, Innovus, Tempus, Conformal, Voltus and Xcelium) and the OpenROAD PSM IR figures are stated in this paper and are not regenerated by the open scripts.

License. The released design files are licensed for non-commercial academic instruction and research; commercial use needs written permission. `picorv32.v` keeps its ISC license, and the SKY130 PDK data keep Apache-2.0. For research collaboration or licensing inquiries about the framework, contact Raisul Islam, raisul@purdue.edu.

References

[1] C. X. Wolf, "PicoRV32 — A Size-Optimized RISC-V CPU," YosysHQ, `https://github.com/YosysHQ/picorv32` (README).

[2] A. Waterman and K. Asanović (eds.), "The RISC-V Instruction Set Manual, Volume I: Unprivileged ISA," RISC-V International.

[3] SkyWater Technology and Google, “SkyWater SKY130 PDK,” `https://github.com/google/skywater-pdk`; open_pdks 1.0.471, `https://github.com/RTimothyEdwards/open_pdks`.